

Carbon

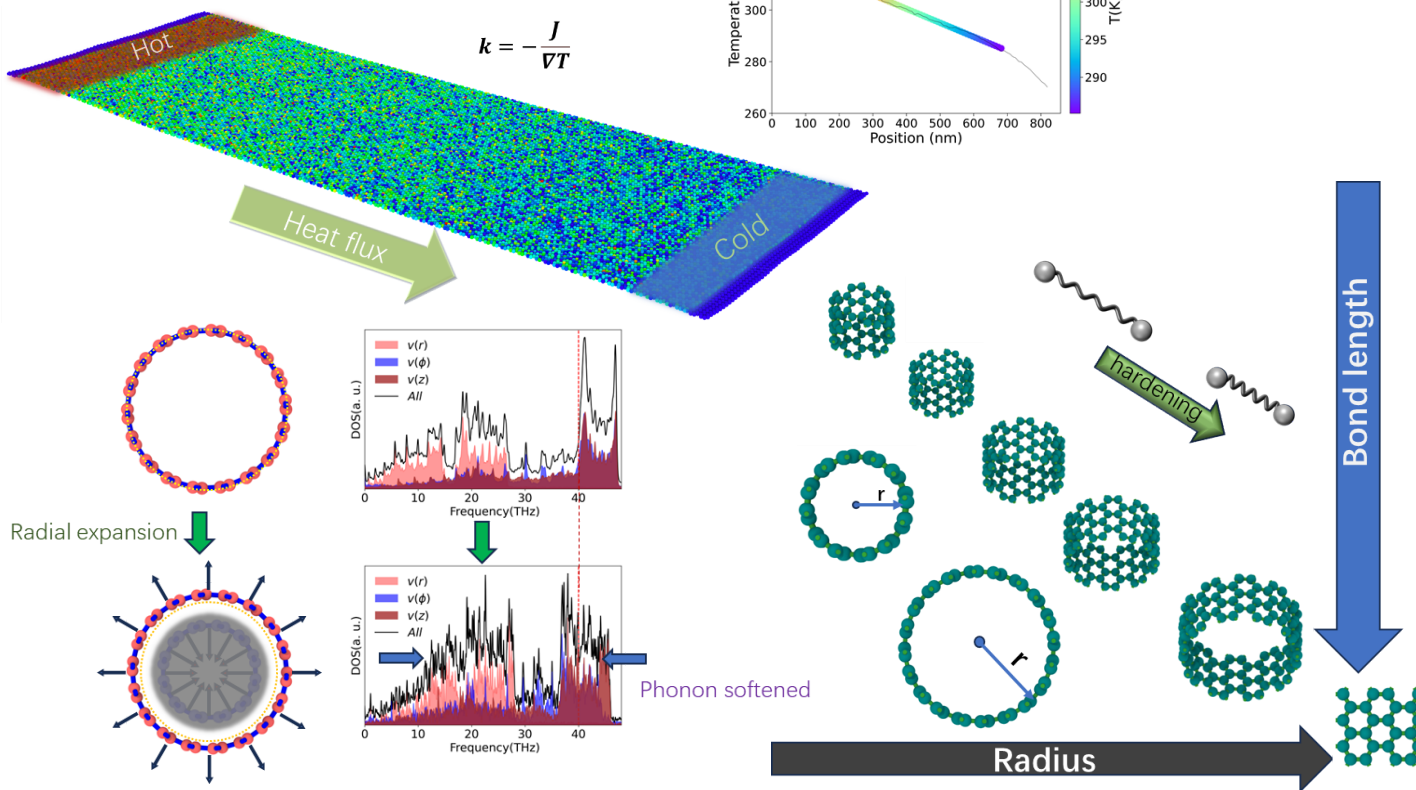
Calculate the Thermal Conductivity of Carbon Nanotubes with NEMD method and Machine Learning Potential --Manuscript Draft--

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Abstract:	Low-dimensional carbon materials, such as graphene and carbon nanotubes, hold great promise for thermal management and nanomechanics. Computational studies on carbon nanotubes (CNTs) face significant challenges due to the inherent computational cost of density functional theory (DFT) methods, which typically have large unit cells with hundreds of atoms. To address this limitation, we employ a modified ReaxFF machine learning interatomic potential (ReaxFF-nn) with parameters optimized for sp²-hybridized carbon to calculate the thermal conductivity of these nanostructures using molecular dynamics (MD) simulations and the non-equilibrium molecular dynamics (NEMD) method. The results reveal that as the diameter of single-walled (SW)CNTs decreases, the C–C bond length elongates from 1.428 to 1.437 Å, leading to phonon softening and approximately 25% reduction in thermal conductivity. In double-walled (DW)CNTs with the interlayer spacing increases from 2.34 to 3.36 Å, resulting in radial expansion that reduces the thermal conductivity. Additionally, the thermal conductivity exhibits a power-law dependence on tube length ($\kappa \propto L^n$), where $n \approx 1$ within the relaxon length scale, and $n < 1$ when the length exceeds the relaxon scale, corresponding to a transition from ballistic to diffusive thermal transport.

Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:



Highlights

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Jia-Hua Liu, Shuo Cui, Feng Guo*, Yu-Shi Wen, Chun-Liang Ji, Heng-Shuai Li, Shou-Xin Cui, Gui-Qing Zhang, Xiao-Chun Wang

- Large-scale and long-time-span non-equilibrium molecular dynamics (NEMD) simulations were performed to determine the thermal conductivity of single-walled carbon nanotubes (SWCNTs) and double-walled carbon nanotubes (DWCNTs) with first-principles precision.
- A machine learning interatomic potential (MLIP), trained using energies and forces from Density Functional Theory calculations, was utilized for the molecular dynamics simulations.
- How the thermal conductivity is affected by the diameter of CNT and the thermal conductivity is reduced in DWCNT compared with their SWCNT counterparts are discussed in depth.

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ABSTRACT

Low-dimensional carbon materials, such as graphene and carbon nanotubes, hold great promise for thermal management and nanomechanics. Computational studies on carbon nanotubes (CNTs) face significant challenges due to the inherent computational cost of density functional theory (DFT) methods, which typically have large unit cells with hundreds of atoms. To address this limitation, we employ a modified ReaxFF machine learning interatomic potential (ReaxFF-*nn*) with parameters optimized for sp^2 -hybridized carbon to calculate the thermal conductivity of these nanostructures using molecular dynamics (MD) simulations and the non-equilibrium molecular dynamics (NEMD) method. The results reveal that as the diameter of single-walled (SW)CNTs decreases, the C–C bond length elongates from 1.428 to 1.437 Å, leading to phonon softening and approximately 25% reduction in thermal conductivity. In double-walled (DW)CNTs with the interlayer spacing increases from 2.34 to 3.36 Å, resulting in radial expansion that reduces the thermal conductivity. Additionally, the thermal conductivity exhibits a power-law dependence on tube length ($\kappa \propto L^n$), where $n \approx 1$ within the relaxon length scale, and $n < 1$ when the length exceeds the relaxon scale, corresponding to a transition from ballistic to diffusive thermal transport.

1. Introduction

The growing demand for efficient heat dissipation in electronic and photonic devices has made high-thermal-conductivity materials essential for sustaining Moore's law[1]. Traditional cooling methods rely mainly on forced convection through metal heat sinks, such as copper or aluminum, whose thermal conductivities are typically limited to a few hundred W/m·K. The next generation of integrated circuits and nanoscale devices requires advanced materials with superior thermal properties to meet the increasing challenges of thermal management.

Over the past decade, low-dimensional materials[2, 3, 4] have attracted substantial research interest as they can enhance the design and efficiency of various advanced systems [5]. Measuring thermal conductivity in low-dimensional materials presents significant experimental challenges. These difficulties arise from inherent technical complexities in characterizing suspended nanostructures[6, 7], limited controllability when systematically varying material dimensions, and lack of standardized protocols for sample quality assessment and measurement techniques. The different experimental conditions and assumptions used in various measurement methods often lead to inaccurate estimates of thermal conductivity [8, 9].

Atomic Molecular Dynamics (MD)[10, 11, 12] simulations offer an attractive alternative. These simulations

allow precise control over the size and structure of low-dimensional nanomaterials and have been used to study how thermal conductivity depends on the length, diameter, elastic deformation, bending, and crystal defects, etc [13, 14]. The primary bottleneck is the lack of interatomic potentials that can accurately and efficiently describe interatomic interactions[15].

Although density functional theory (DFT) calculations [16], which involve solving the Schrödinger equation, provide precise descriptions of interatomic interactions, they are generally restricted to systems with only a few hundred atoms and time scales of picoseconds[17]. In contrast, simulations using empirical potentials with explicit functional forms are significantly faster, by several orders of magnitude, but suffer from lower accuracy and limited transferability[18, 19].

Machine Learning Interatomic Potentials (MLIPs)[20] are emerging as new tools to bridge the gap between DFT and empirical potentials. In recent years, significant efforts have been made to construct interatomic potentials using ML models such as artificial neural networks[21], Behler–Parrinello neural-network potential (BPNNP)[22], spectral neighbor analysis potentials (SNAP)[23], Moment Tensor Potentials (MTP)[24], Deep Potential (DP)[25], Atomic Cluster Expansion (ACE)[26], Reactive Force Field with Neural Networks (ReaxFF-*nn*)[27], Neuroevolution Potential (NEP)[28], and Gaussian Approximation Potential (GAP)[29], offering enhanced predictive capability for complex systems. MLIPs have been demonstrated to predict structural, physical, mechanical, chemical, and thermodynamic properties with DFT-level accuracy. At the same time,

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they can reduce computational costs by several orders of magnitude, greatly expanding the spatial and temporal scales of atomic simulations[30, 31, 32].

The Non-Equilibrium Molecular Dynamics (NEMD)[33, 34, 35] method is a simulation technique for characterizing material properties under non-equilibrium conditions. This approach computes the thermal conductivity by setting cold and hot regions within the material, thereby inducing a temperature gradient. The material's thermal conductivity can be calculated from the temperature gradient based on Fourier's law. This method better approximates the non-equilibrium conditions of actual applications, providing a more realistic simulation of material transport behavior under working conditions. Compared to Equilibrium Molecular Dynamics (EMD)[36, 37], NEMD allows direct measurement of heat flux density, avoiding the need to fit autocorrelation functions and thus reducing errors. The NEMD method has significant advantages in handling strong non-equilibrium states and complex dynamic behaviors[38, 39].

Graphene[40, 41] is a two-dimensional material composed of a single layer of carbon atoms arranged in a hexagonal honeycomb lattice. Graphene exhibits extremely high thermal conductivity, ranging from 3000 to 5000 W/mK[7, 42, 43] at room temperature. This is attributed to the strong covalent bonds between carbon atoms and the mean free paths of long phonons.

Carbon nanotubes (CNTs)[44] are one-dimensional nanomaterials with a hollow tubular structure formed by rolling graphene sheets. Depending on the rolling method, CNTs can be classified as single-walled carbon nanotubes (SWCNTs) [45] or multi-walled carbon nanotubes (MWCNTs)[46]. Despite extensive research on the thermal conductivity of carbon nanotubes and graphene, there remain controversies regarding their thermal conductivity values[47, 48].

This study employs non-equilibrium molecular dynamics (NEMD) simulations with the ReaxFF-*nn* machine learning potential[27, 49] to investigate the thermal conductivity of carbon-based nanomaterials. The effects of diameter and chirality on the thermal conductivity of carbon nanotubes and the impact of length and temperature on the thermal conductivity of monolayer graphene sheets are discussed. Specifically, the dependence of length on the thermal conductivity of SWCNTs and DWCNTs is examined. The simulation results show good agreement with previous experimental and theoretical findings.

2. Methodology

2.1. Molecular Dynamics Methods and Material Modeling Details

This study employs non-equilibrium molecular dynamics (NEMD) simulations using the ReaxFF-*nn* potential. All molecular dynamics simulations were conducted using the Large-scale Atomic/Molecular Massively Parallel Simulator (LAMMPS)[50, 51]. A time step of 0.1 fs was adopted. The system was first relaxed under NPT and NVT ensembles for 0.5 ns each to fully eliminate internal stress, followed

by NVE ensemble simulations of up to 5 ns depending on the system length. The materials studied in this work include single-layer graphene, armchair carbon nanotubes ranging from (4,4) to (10,10), double-walled carbon nanotubes (DWCNTs) with normal interlayer spacing, such as (4,4)@(9,9), and DWCNTs with abnormal interlayer spacing, such as (6,6)@(9,9) and (5,5)@(9,9). The NEMD simulation setup is illustrated in Figure 1(a), where the simulation length is denoted as L . Periodic boundary conditions were applied perpendicular to the heat transfer direction, while free boundary conditions were imposed along the heat transfer axis. Fixed layers were positioned at both ends to constrain atomic motion. Previous studies have demonstrated that the Langevin thermostat is more effective and reliable than the Nosé-Hoover (chain)[52, 53] method for local temperature control in NEMD simulations. Consequently, Langevin thermostats were applied to cold and hot regions to establish heat sinks, with the thermostat length and damping parameter adjusted according to the simulation length[54]. All structural visualizations were rendered using Ovito[55].

2.2. Thermodynamic Property Calculations

This study investigates the thermal conductivity of single-layer graphene and various carbon nanotube structures as a function of length using Fourier's law

$$\kappa = -\frac{J}{\nabla T}, \quad (1)$$

where J represents the heat flux density in the material, and the temperature gradient ∇T is extracted by performing piecewise linear regression on the temperature distribution along the transport direction. The calculation of the total heat flux and the temperature gradient has been clearly explained in previous studies[56, 57]. In the calculation, the cross-sectional area A of single-layer graphene is expressed as $A = W \times h$, where W is the width of the graphene sample and h is the van der Waals interaction distance, and in this study, $h = 0.34 \text{ nm}$ is adopted.

For single-walled carbon nanotubes, the cross-sectional area S is defined as $S = \pi \cdot h \cdot D$ (i.e., the tube is treated as a cylindrical structure), where D is the nanotube's diameter. To facilitate comparison with single-walled carbon nanotubes, as shown in Figure 1(b), the cross-sectional area of multi-walled carbon nanotubes is defined as the sum of the cross-sectional areas of two nested single-walled carbon nanotubes, regardless of whether the interwall distance exceeds 0.34 nm .

The NEMD method requires the collection of heat flux and temperature data for thermal conductivity calculations after the system reaches a steady state. Therefore, determining when the system has reached steady state is crucial. Based on previous researches, we assess the system's steady state from three aspects as presented in Figure 2: (i) the first is a stable temperature gradient, with the temperature averaged over a 100 ps sampling period; (ii) the second is stable heat flux, recorded as the energy exchanged by the heat sinks over the same 100 ps period; (iii) the third is a uniform and stable temperature distribution, with temperature data

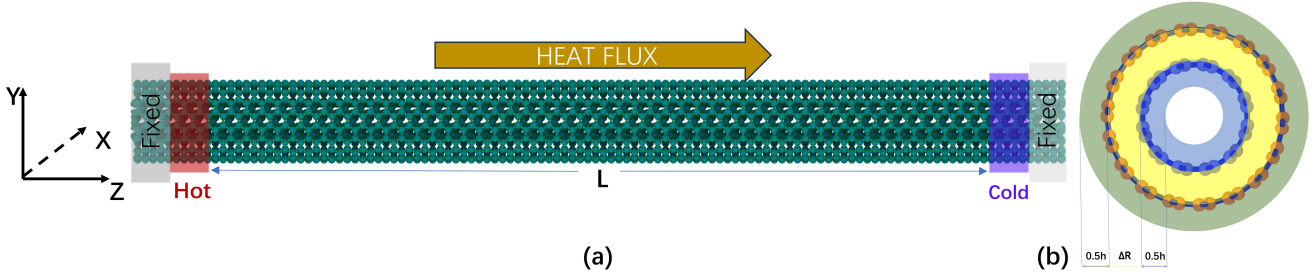


Figure 1: (a) A schematic illustration of heat transfer, where the direction of heat flow in carbon nanotubes and graphene is depicted. (b) For the calculation of the heat flux density J , the cross-sectional area A is defined as comprising the red, yellow, and blue regions. The red region is defined as the area obtained by extending the outer wall of a double-walled carbon nanotube outward by $0.5h$ (where $h = 0.34 \text{ nm}$), the yellow region represents the area between the cross-sectional circles of the inner and outer walls, and the blue region is defined as the area obtained by extending the inner wall inward by $0.5h$.

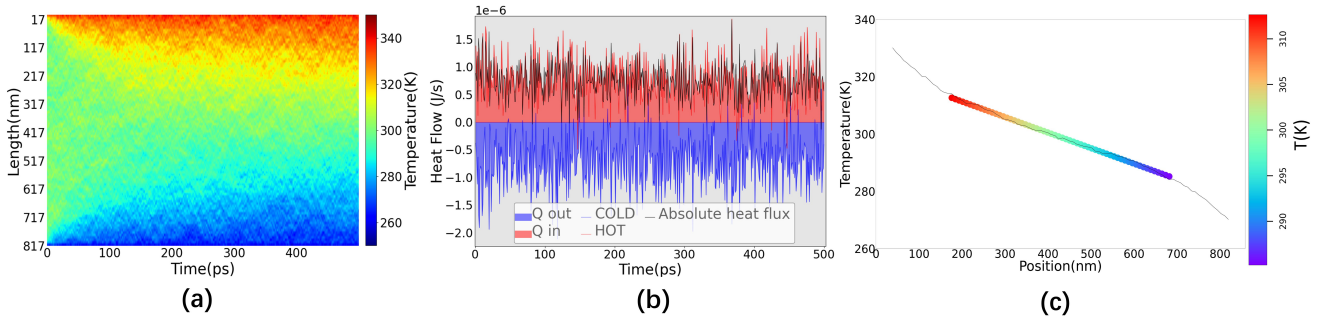


Figure 2: (a) The temporal evolution of the temperature distribution along the heat transfer direction in a 853 nm graphene sample is presented for the first 500 ps, with the y-axis representing different regions along the heat conduction direction. (b) Heat flux as a function of time. Q_{in} is defined as the energy absorbed by the cold region during the calculation, Q_{out} is defined as the energy lost by the hot region, J_{cold} is used to represent the heat flux in the cold region (positive, due to energy absorption), and J_{hot} is used to represent the heat flux in the hot region (negative, due to energy loss). The absolute heat flux is defined as the average of the absolute values of the heat fluxes in the cold and hot regions. After system stabilization, this average is determined to be the long-term average of the heat flux J in Equation (1). (c) Fitted line of the stabilized temperature gradient. The black line connects the average temperatures across different regions along the heat transport direction, and a linear fit is performed on the central region with a relatively uniform temperature gradient, as represented by the colored dots.

recorded once the system enters the NVE ensemble. The system must satisfy all three conditions simultaneously to obtain reliable thermal conductivity[58].

2.3. Phonon Calculations

Phonon analysis is widely recognized as an effective way to understand thermal conductivity mechanisms. We calculated and analyzed the phonon properties of the simulated materials to gain a deeper understanding of the fundamental physical mechanisms behind thermal conduction in these materials.

In the present work, first-principles calculations were performed using the SIESTA software[59]. The Phonopy package[60] was used to calculate the phonon dispersion relations of the materials, where the force constants were obtained using the finite displacement method. The structures were optimized using the ReaxFF-*nn* force field before the calculations. For single-layer graphene, the unit cell was expanded to $8 \times 8 \times 1$ supercell size, and for all carbon nanotubes, the unit cell was expanded to $1 \times 1 \times 8$ along the Z direction.

Additionally, **Alamode** software [61] was used to calculate the anharmonic properties of phonons in graphene and evaluate its thermal conductivity, with the latter determined via Equation (2). The phonon density of states (DOS) was extracted from the atomic trajectories using Equation (3), where i denotes the atomic index. The equipartition theorem is commonly employed in MD simulations. However, this approximation is only valid at high temperatures, specifically when the temperature exceeds the Debye temperature[62]. To address this limitation, we estimated the quantum correction factor[63] and found its contribution to be minor. Consequently, quantum corrections were not applied to the final results. Detailed calculations of the quantum correction are provided in Supplementary Fig. 1.

$$\kappa_L^{\alpha\beta}(T) = \frac{1}{NV} \sum_q C_q(T) v_q^\alpha(T) v_q^\beta(T) \tau_q(T) \quad (2)$$

$$D(\omega) = \int_0^\infty \frac{\sum_i \langle v_i(t_0) v_i(t_0 + \tau) \rangle}{\sum_i \langle v_i(t_0) v_i(t_0) \rangle} e^{2\pi i \omega \tau} d\tau \quad (3)$$

2.4. Interatomic Interaction Potential – ReaxFF-*nn*

Empirical interatomic potentials typically express the system energy as a function of atomic positions. Notable examples include the Tersoff [64] and embedded-atom method (EAM) potentials[65]. While effective for modeling systems without bond formation or dissociation, these nonreactive potentials are inadequate for simulating chemical reactions. In contrast, the ReaxFF force field[66] incorporates bond order terms to describe dynamic bonding behavior, including bond breaking and formation. The bond order is derived from empirical functions based on changes in interatomic distances.

The total energy expression of ReaxFF is given in Equation

$$E_{bond_{ij}}^{ReaxFF} = D_e^\sigma \cdot BO_{ij}^\sigma \cdot \exp[p_{be1}(1 - BO_{ij}^\sigma)^{p_{be2}}] - D_e^\pi \cdot BO_{ij}^\pi - D_e^{\pi\pi} \cdot BO_{ij}^{\pi\pi}, \quad (4)$$

$$E_{bond_{ij}}^{ReaxFF-nn} = D_e^\sigma \cdot f_{nn}(BO_{ij}^\sigma, BO_{ij}^\pi, BO_{ij}^{\pi\pi}). \quad (5)$$

The bond energy term, E_{bond} , as in Equation (4), is a continuous function of interatomic distances and represents the energy associated with bond formation. The angle and torsion terms, E_{angle} and E_{tors} , describe strain energies arising from deviations in bond angles and torsions, respectively. E_{over} penalizes atomic overcoordination, while E_{coul} and E_{vdW} represent electrostatic and van der Waals interactions. In previous work, we proposed a new potential model called ReaxFF-*nn* by modify the ReaxFF bond order and energy formulation, as shown in Equation (5), the short-range interaction term E_{bond} is calculated by a simple full connected neural network, whereas the long-range interactions, such as van der Waals and Coulomb terms, retain the original ReaxFF formalism. While sharing fundamental similarities with the original ReaxFF formalism, the ReaxFF-*nn* demonstrates significantly enhanced computational performance.

2.5. Interatomic Potential Train

Prior to MD simulations, the potential model ReaxFF-*nn* was trained against the energies and forces derived from DFT-MD simulations. The corresponding error metrics for energies and forces are presented in Supplementary Fig. 2. The DFT calculations in this work were performed using the SIESTA[59] software with the Perdew-Burke-Ernzerhof (PBE) exchange-correlation functional and a double- ζ polarized basis set. Training data are generated through three distinct DFT computational approaches: (i) the DFT-MD simulations at varied temperatures and pressures; (ii) the atomic structure scaling, for example, scaling the supercell of graphene or CNTs; (iii) configurations sampled from classical MD simulations. In the last step, the classical MD simulations were conducted with parameters trained with existing data. The classical MD will produce a large number of atomic structures. To prevent dataset redundancy, structural subsets are systematically sampled from MD trajectories via

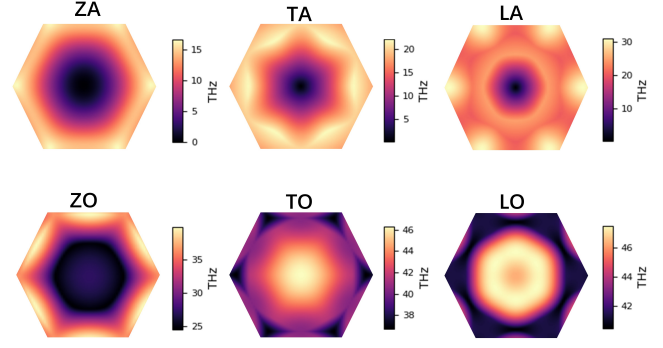


Figure 3: The six different phonon dispersion curves of graphene within the hexagonal two-dimensional Brillouin zone.

random selection or extraction of configurations exhibiting anomalous energies or unphysical bond lengths. This process is iterated until the interatomic potential model meets target performance criteria. The training of the potential is done with our I-ReaxFF[67] package.

2.6. Phonon Spectra

Phonon spectra are highly sensitive to the parameters of an interatomic potential, making them a critical benchmark for accuracy. Furthermore, phonon dispersion relations offer a convenient means of validation, as they can be directly compared with results from both first-principles calculations and experimental measurements. To evaluate the accuracy of our potential in describing phonon dispersion, we computed the phonon spectrum across the entire first Brillouin zone of graphene. The phonon dispersion relations obtained in this Brillouin zone are shown in Figure 3.

The Brillouin zone of a carbon nanotube is derived from folding that of graphene, as illustrated in Supplementary Fig. 3. The allowed wavevectors are detailed in the Supplementary Information. We further compared our results with those obtained from first-principles calculations. Figure 4 presents the phonon dispersion curves of a (5,5) SWCNT along its one-dimensional Brillouin zone, formed by folding the graphene Brillouin zone. Panel (a) displays the results obtained using Phonopy with the ReaxFF-*nn* potential, while panel (b) shows the force constants extracted from SIESTA calculations. The illustration shows that the ReaxFF-*nn* potential calculation provides a good match to the low-frequency region of the phonon spectrum. However, its ability to resolve adjacent phonon branches is somewhat inferior compared to the SIESTA results. Additionally, the high-frequency optical branches predicted by ReaxFF-*nn* are generally lower than those from SIESTA, with slight discrepancies in branch positions. Nonetheless, these optical modes contribute negligibly to thermal conductivity.

2.7. Thermal Conductivity Calculation

Phonon dispersion relations are generally calculable from second-order force constants. However, to obtain a physically realistic and finite thermal conductivity, third- and fourth-order anharmonic interactions must be taken

into account. As thermal conductivity represents a critical material property with broad engineering applications, its rigorous investigation remains essential for ensuring reliability in practical implementations. Therefore, the thermal conductivity of graphene is calculated using two approaches: one based on higher-order force constants and the other using non-equilibrium molecular dynamics (NEMD) with Fourier's law. Figure 1 compares the thermal conductivity of graphene computed via NEMD at different sample lengths with results obtained using the force-constant approach. It is reported that the relaxation time approximation (RTA)[48, 68] method (e.g., Eq. 3) significantly underestimates the thermal conductivity of materials with weak anharmonicity, such as graphene[69], the computed thermal conductivity values were normalized to those at the largest simulated size (853 nm). This normalization facilitates the comparison of relative trends while mitigating the inherent limitations of the RTA approach in capturing anharmonic effects. Figure 6 quantifies the contributions of phonons at different frequencies to the overall thermal conductivity, revealing that low-frequency phonons dominate heat transport while optical modes above 20 THz contribute less than 20%.

3. Results and Discussions

3.1. Size Effect

Unlike bulk materials, the thermal conductivity of low-dimensional materials exhibits strong size dependence, a phenomenon known as the size effect[70]. This dependence originates from enhanced phonon-boundary scattering in confined geometries. Previous studies have demonstrated that as the length of a graphene sample increases, phonon transport transitions from ballistic to quasi-ballistic and ultimately to diffusive mode[39, 71]. Specifically, transport is ballistic when the sample length is significantly smaller than the phonon mean free path (MFP) [72], quasi-ballistic when the length is comparable to the MFP, and diffusive when the length substantially exceeds the MFP. These distinct transport regimes give rise to different trends in thermal conductivity.

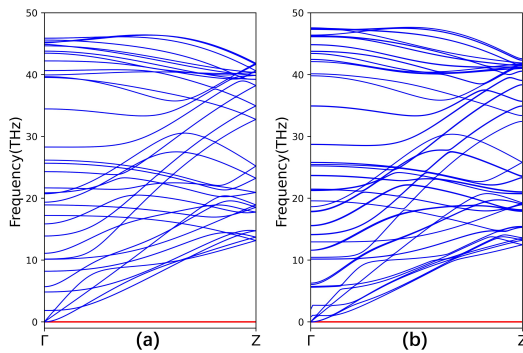


Figure 4: Phonon dispersion relations of carbon nanotubes along the one-dimensional Brillouin zone: (a) results obtained using ReaxFF-nn potential, (b) results from SIESTA first-principles calculations.

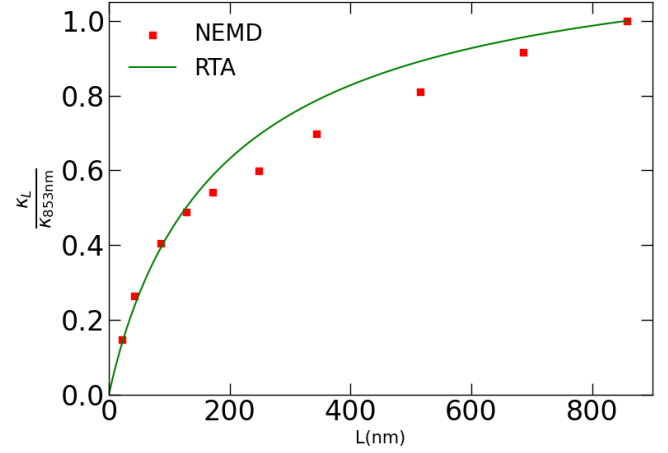


Figure 5: Comparison of normalized thermal conductivity at 850 nm as predicted by NEMD simulations (red squared points) and RTA method calculations with ALAMODE package.

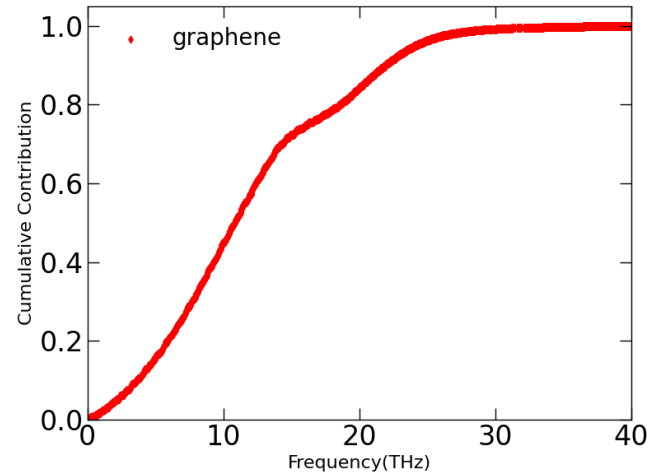


Figure 6: Normalized cumulative thermal conductivity versus frequency.

The size effect from both relaxon and phonon perspectives has been analyzed following the theoretical framework by Andrea Cepellotti et al.[73]. Non-equilibrium lattice heat transport cannot be fully described by the properties of individual phonons, as scattering tightly couples phonons of different wavevectors and branches [74]. In other words, heat transport properties must be described in terms of collective phonon populations. Relaxons are collective excitations composed of linear combinations of non-equilibrium phonon populations. Compared to phonons, relaxons exhibit shorter relaxation times and lower group velocities, resulting in a shorter MFP.

In the ballistic transport regime, thermal conductivity increases linearly with sample length, indicating a constant thermal conductance. However, as shown in Fig. 7(d), the

thermal conductivity of graphene begins to decrease at approximately 86 nm, signifying the boundary of the ballistic regime. This transition aligns with the relaxon MFP distribution calculated by Cepellotti et al., which shows sparsity around 100 nm. The end of the linear growth is generally regarded as corresponding to the effective phonon MFP, indicating the boundary of the ballistic regime[38]. It should be noted that heat transport is described using relaxons instead of phonons, with the system size, after exceeding the maximum MFP of relaxons, continuing to increase and gradually approaching the phonon MFP limit. During this stage, the thermal conductivity remains unsaturated, in a regime termed quasi-ballistic transport.

$$\kappa(L)^{-1} = \kappa_{\infty}^{-1} (1 + \lambda/L) \quad (6)$$

The quasiballistic thermal conductivity can be described by Equation (6) [75], where L represents the sample length, λ is the effective phonon MFP (distinct from the MFP of individual phonons, as it accounts for collective phonon behavior), and κ_{∞} represents the saturated thermal conductivity in the diffusive limit. By fitting the data points from 245 to 853 nm, the diffusive limit is $\kappa_{\infty} = 2200 \text{ W/mK}$ at $\lambda = 420 \text{ nm}$. This value is close to the PBE functional calculation ($\kappa = 1936.156 \text{ W/mK}$) reported by Qin[76].

In the ballistic regime, the increasing sample length enables relaxons with longer MFPs to contribute to heat transport. In the quasi-ballistic regime, the effective MFP increases with system size due to phonon-phonon interactions. The effective MFP of the phonon ensemble increases with the system size because boundary scattering allows phonons that previously scattered at the boundary to interact with each other. From a thermodynamic perspective, high-energy phonons are more likely to undergo Umklapp (U)[77] scattering than low-energy phonons. These high-energy phonons interact with phonons that would otherwise scatter at the boundaries, increasing the ratio of Normal-to-Umklapp scattering. As a result, the average distance between successive Umklapp scattering events increases the phonon MFP and delays the saturation of thermal conductivity. As the system size grows, phonons that previously underwent boundary scattering have more opportunity to participate in phonon-phonon interactions. This ultimately leads to a thermal conductivity plateau and goes to the diffusive regime boundary. This phenomenon has also been experimentally observed[71].

Quasi-one-dimensional materials, such as carbon nanotubes (CNTs), also exhibit size-dependent thermal conductivity like graphene. Previous studies demonstrate that ballistic thermal transport dominates in carbon nanotubes (CNTs) when the tube length is shorter than the maximum phonon MFP, exhibiting linear dependence of thermal conductivity on length, i.e., $\kappa \propto L$. In the quasi-ballistic regime, the thermal conductivity scales sublinearly with length, which can be expressed as $\kappa \propto L^n$ ($0 < n < 1$)[42]. As shown in Fig. 7(b), compared to relaxon MFPs in graphene, the average relaxon MFPs for SWNT (6,6) and SWNT (9,9) are 184 and 161 nm, respectively. Moreover, the relationship

between the effective MFP phonon and the macroscopic thermal conductivity in CNTs is fitted using Equation (3). As shown in Fig. 7(a), the results show that the effective phonon MFPs in CNTs are shorter than those in graphene. Additionally, among CNTs with different diameters, the effective phonon MFP of the (9,9) nanotube (88 nm) is longer than that of the (6,6) nanotube (79 nm). This difference may be attributed to the constrained phonon interactions resulting from the curvature introduced by rolling graphene into nanotubes.

For smaller-sized CNTs, the amplitude of atomic vibrations is greater, which can be directly observed in the molecular dynamics trajectories. The thermal conductivity is reduced by the axial curvature variations. Although prior studies have addressed the effect of axial curvature on thermal conductivity[78], establishing precise quantitative descriptions remains challenging.

3.2. Diameter-Dependent Thermal Conductivity in CNTs

By comparing armchair single-walled carbon nanotubes (SWCNTs) of different diameters at a fixed length, we found that the thermal conductivity of SWCNTs is lower than that of monolayer graphene at the same length. However, as the diameter increases, the thermal conductivity gradually approaches the value of graphene. Notably, as observed in recent experimental studies[79], the rate of increase in thermal conductivity decreases with nanotube diameters.

We attribute this trend to two main factors. Firstly, analysis of the phonon density of states (PDOS) shows that increasing the diameter induces a blue shift in the high-frequency modes and a red shift in the low-frequency modes, corresponding to the phonon hardening effect as reported by Kodama[80], Parrish[81], and Xiao[82]. Such shifts reduce the probability of Umklapp (U) scattering, thereby improving the thermal conductivity of larger-diameter carbon nanotubes. Secondly, changes in relaxation time also play a role. Previous studies have shown that increased curvature can reduce the phonon relaxation time[83]. This effect can be qualitatively captured by the relation $\tau = (1 - 0.5h/D)$, where τ denotes the phonon relaxation time, h is the effective thickness of graphene (0.34 nm), and D is the nanotube diameter. The thermal conductivity of carbon nanotubes with different diameters was fitted by the relative phonon density of states of the acoustic branches and the changes in phonon relaxation time, as described in Equations (7) and (8). In Equation (8), κ_g is defined as the simulated thermal conductivity of graphene obtained from non-equilibrium molecular dynamics simulations. The quantities $G(20)_g$ and $G(20)_{cnt,z}$ represent the cumulative relative phonon density of states for graphene and carbon nanotubes, respectively, calculated up to 20 THz using Equation (7). The fitting results and the comparison with the NEMD simulation data are presented in Figure 9b. Furthermore, the curvature of CNTs alters the C-C bond lengths compared with graphene, and can be evidenced by the radial distribution function (RDF) analysis[84]. This aspect will be discussed in detail

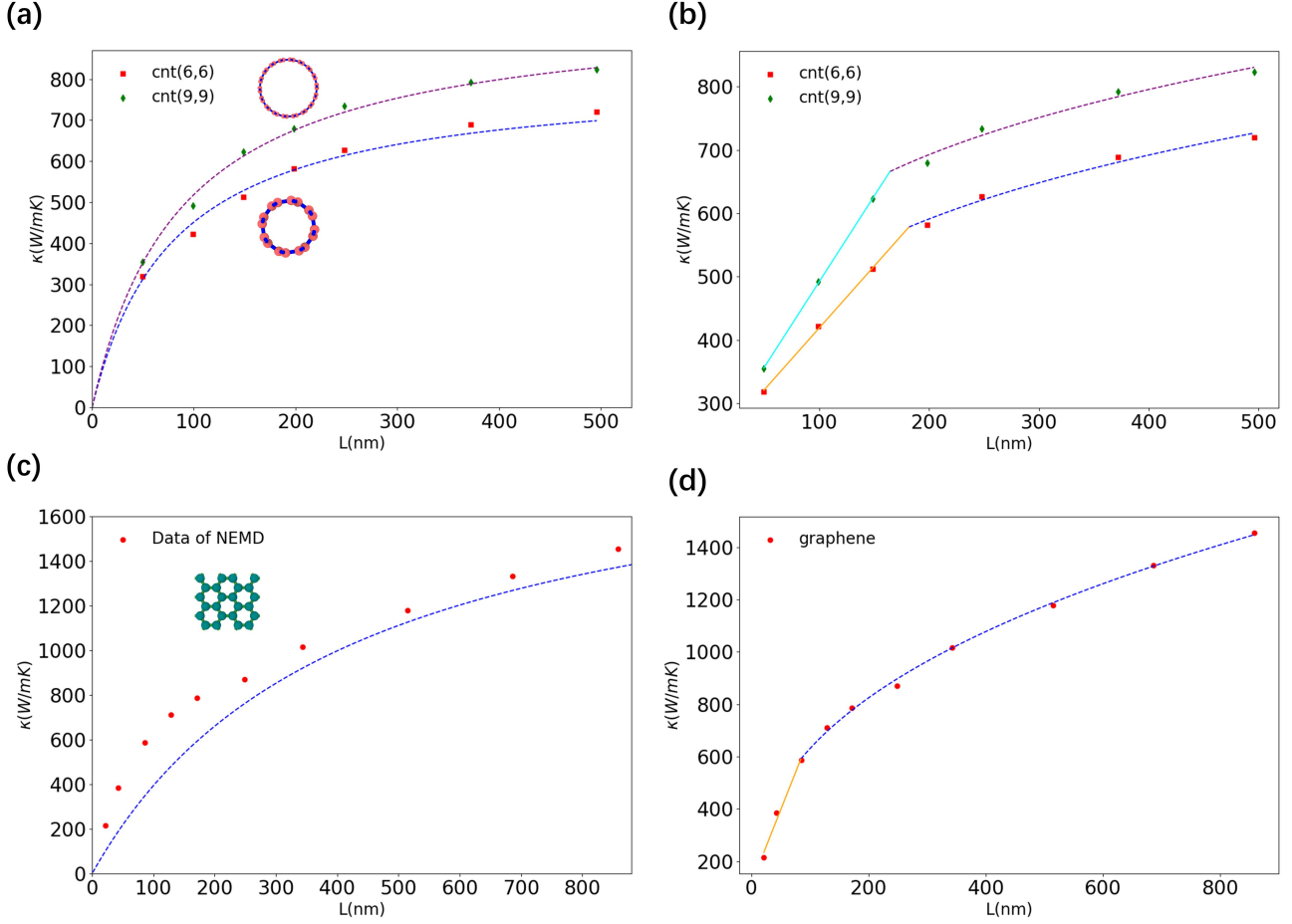


Figure 7: (a) The comparison of thermal conductivity of CNT(6,6) and CNT(9,9) at different sizes, dotted symbols represent data obtained from simulations, and dashed lines are used to represent results fitted according to Equation (6). (b) The results of alternative fitting methods for the data in (a), solid lines represent linear fitting results, and dashed lines represent results fitted according to a power law. (c) The comparison of thermal conductivity of graphene at different sizes, dotted symbols represent data obtained from simulations, and dashed lines represent results fitted according to Equation (6). (d) The results of alternative fitting methods for the data in (c), solid lines represent linear fitting results, and dashed lines represent results fitted according to a logarithmic function.

in the following section.

$$G(cum) = \frac{\int_0^{cumTHz} dos(\omega) d\omega}{\int_0^{\infty} dos(\omega) d\omega}$$

$$\kappa_{fit} = \kappa_g \frac{G(20)_{cnt}(1 - 0.5h/D))}{G(20)_g}$$

3.3. Phonon Density Of States

Figure 11 shows the low-frequency phonon density of states (PDOS) for carbon nanotubes ranging from SWNT(7,7) to SWNT(10,10). As the diameter of carbon nanotubes increases, the peak in radial phonon density of states (PDOS) corresponding to the radial breathing mode shifts to lower frequencies. The first low-frequency peak moves from 2.45 THz in SWNT(4,4) to 0.64 THz in SWNT(10,10). Since monolayer graphene lacks a pronounced peak in this low-frequency region, no direct comparison is made. The nonzero

intensity near 0 THz arises from the fixed boundary conditions imposed at both ends[48]. In contrast, the high-frequency peaks in the total phonon density of states (PDOS) exhibit a clear shift towards higher frequencies as the diameter increases. This behavior is prominently observed in Figure 8, where the red straight line clearly shows the peaks moving from below 40 THz to above 40 THz. Since phonons responsible for thermal transport primarily lie below 20 THz[85], we used the relaxation time approximation discussed in the previous section to fit the thermal conductivity of nanotubes with different diameters using Equations 10 and 11, and compared the results with our simulation data, as shown in Figure 8. A comparison of contributions to phonon density of states from different vibrational modes reveals that the radial vibrations in carbon nanotubes resemble the out-of-plane (Z-direction) vibrations in graphene, which stems from the fact that CNTs are essentially rolled-up graphene sheets. Likewise, the twisting and axial vibrations

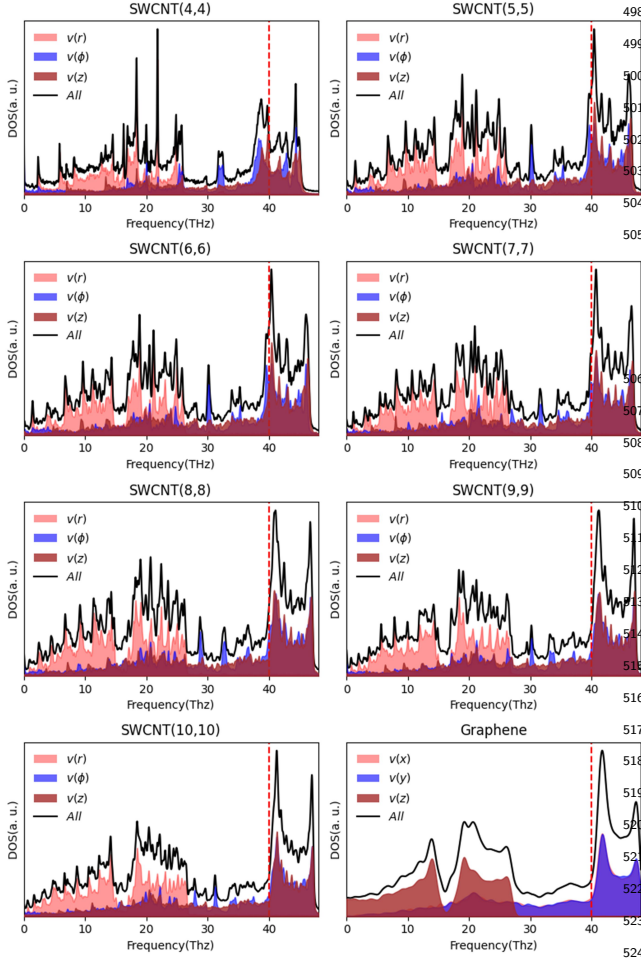


Figure 8: The phonon density of states for carbon nanotubes with varying diameters and graphene is compared, where a black line is used to represent the total phonon density of states, and filled areas of different colors are used to represent the density of states in specific directions. In carbon nanotubes, orange filled areas are used to represent the radial density of states, blue filled areas are used to represent the torsional density of states, and red filled areas are used to represent the axial density of states; in graphene, orange, blue, and red filled areas are used to represent the density of states in the x , y , and z directions, respectively.

in CNTs correspond to the in-plane (X and Y directions) vibrations in graphene. Details of the velocity component calculations are presented in the Supplementary Fig. 4. It is also observed that out-of-plane vibrations constitute the dominant contribution to the low-frequency PDOS. Previous studies demonstrate that this region is more sensitive to the nanotube diameter[86]. Enlarged nanotube diameter elevates low-frequency phonon state density, consequently enhancing the thermal conductivity.

3.4. RDF

The RDF analysis by equation(12) of carbon nanotubes with varying diameters reveals that the position of the first peak, which represents the nearest-neighbor bond length,

decreases gradually as the tube diameter increases. This indicates that bond lengths vary with diameter. Zhang's [87] studies have shown that when graphene is rolled into a nanotube, the original uniform bond length of 1.421 Å in graphene splits into three distinct values: d_1 , d_2 , and d_3 . In armchair nanotubes, $d_2 = d_3$. However, due to thermal fluctuations, only one peak corresponding to the bond length could be identified in the RDF analysis.

$$g(r) = \frac{1}{\rho_0} \cdot \frac{\Delta N(r)}{4\pi r^2 \Delta r} \quad (9)$$

The full width at half maximum (FWHM) of the RDF peak reflects the crystal stiffness. Figure 9a demonstrates that increasing the nanotube diameter leads to a gradual reduction in both the C–C bond length and the corresponding FWHM, suggesting enhanced structural stability. A narrower peak signifies higher stiffness, which promotes harmonic vibrations and reduces phonon scattering. Consequently, this leads to enhanced thermal conductivity[88]. These findings align with the thermal conductivity measurements under high pressure reported by Zhou et al.[89]. In contrast, as the tube diameter decreases, the RDF peak broadens relative to that of graphene, indicating a reduction in the stiffness. This increased anharmonicity enhances phonon scattering, thereby lowering the thermal conductivity in smaller-diameter nanotubes. Furthermore, variations in the nanotube's axial length can also influence bond lengths and RDF peak widths. Analysis of the phonon density of states (PDOS) reveals a trend analogous to that observed in studies of strained graphene by Kumar et al.[90], where tensile strain shifts phonon modes towards intermediate frequencies[91]. This suggests that the process of rolling graphene into a nanotube introduces tensile strains, particularly pronounced in smaller-diameter tubes. As evidenced by the shift in PDOS towards intermediate frequencies, this strain contributes to a reduction in thermal conductivity.

Similar to the observed trend in thermal conductivity, the bond length exhibits the most significant change between CNT(4,4) and CNT(5,5), beyond which the rate of change diminishes. As the nanotube diameter increases, both metrics gradually approach the values of graphene.

3.5. Double-Walled Carbon Nanotubes

Figure 10 reveals significant deviations in bond lengths between double-walled carbon nanotubes (DWCNTs) and their corresponding single-walled counterparts. Among the configurations studied, the 55@99 DWCNT shows the smallest change, approximately 20 femtometers (fm) for inner and outer walls, and achieves the highest thermal conductivity. In contrast, the 44@99 DWCNT shows a contraction of 120 fm in the inner wall and an expansion of 50 fm in the outer wall. The most pronounced bond length alterations occur in the 66@99 structure, with an inner wall contraction of 2100 fm and an outer wall expansion of 2250 fm , resulting in the lowest thermal conductivity. These variations are attributed to excessive inter-wall strain. This strain significantly reduces phonon transport efficiency, causing

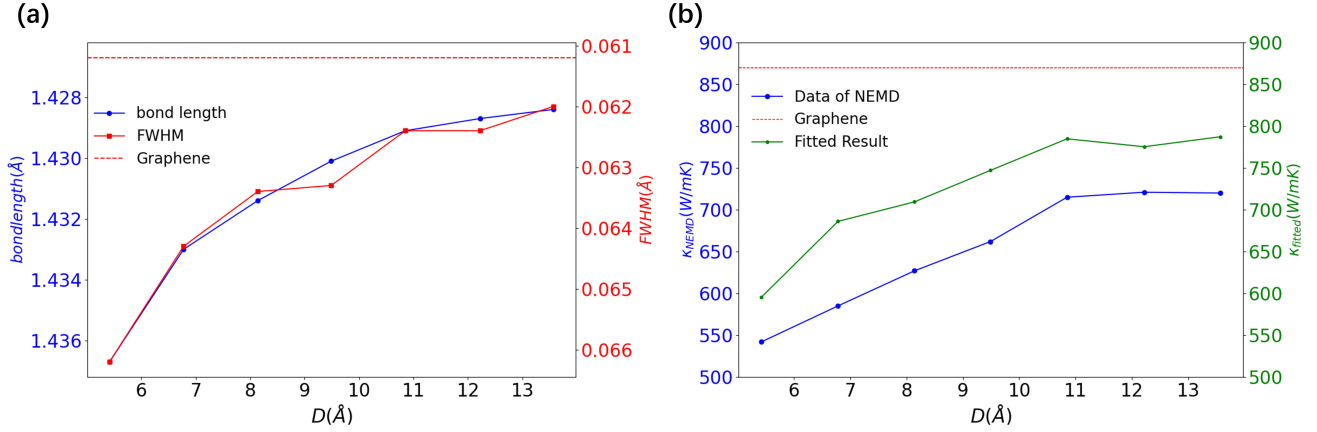


Figure 9: (a) The position of the first nearest peak and the full width at half maximum (FWHM) of carbon nanotubes with different diameters. (b) Comparison of the thermal conductivity calculation results of carbon nanotubes with different diameters and the fitted results by Equation (8).

the thermal conductivities of all multi-walled nanotubes to fall below that of the outer tube of the corresponding (9,9) single-walled CNT[92]. The differences in the phonon density of states induced by strain in graphene are detailed in Fig. 5 and Fig. 6 of the supplementary materials.

Furthermore, the analysis of molecular dynamics trajectories reveals that multi-walled carbon nanotubes (MWCNTs) exhibit smaller vibrational amplitudes than their SWCNTs. Therefore, the reduction in thermal conductivity due to curvature effects should be less significant in MWCNTs than in SWCNTs. However, simulation results demonstrate that MWCNTs of the equivalent size exhibit lower thermal conductivity than SWCNTs. This apparent contradiction suggests additional thermal transport suppression mechanisms in MWCNTs. One reason may be the phonon-phonon scattering. Previous studies have shown that van der Waals (vdW) interlayer interactions enhance phonon-phonon scattering, thereby reducing the thermal conductivity of MWCNTs. Additionally, a recent study by Sun et al.[93] discussed the uncharted phonon modes in DWCNTs arising from degenerate coupling of transverse acoustic modes between the inner and outer tubes. The impact of these newly identified modes on thermal conductivity must also be considered.

4. Conclusions

In summary, by employing the ReaxFF-*nn* machine-learning potential, this study calculated the thermal conductivity of graphene, single-walled carbon nanotubes (SWCNT), and double-walled carbon nanotubes (DWCNT) via the relaxation time approximation and Non-equilibrium molecular dynamics simulations, analyzed their size and diameter dependence, and discussed the thermal conductivity of DWCNT in depth.

These results reveal that as the system size increases, phonons should be treated as an interacting collective rather than independent ballistic carriers, which is different from previous studies based on ballistic transport models. Our

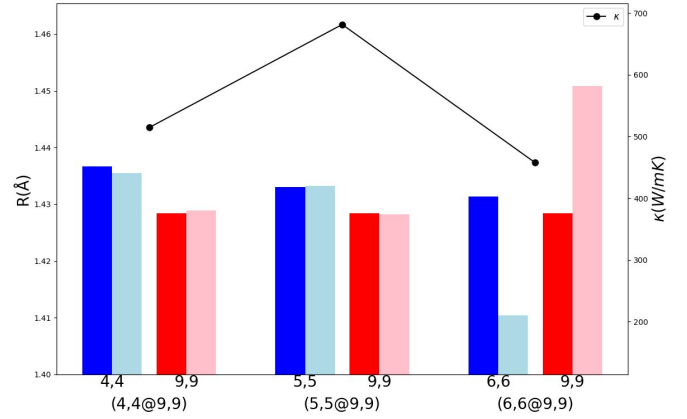


Figure 10: For DWCNTs with different interlayer spacings, variations in their inner and outer diameters and the corresponding changes in thermal conductivities were compared.

results demonstrate that κ of CNT increases with diameter, asymptotically approaching the value for graphene. This trend is attributed to curvature-induced modifications of phonon relaxation times, which can be qualitatively described by the relationship $\tau = (1 - 0.5h/D)$. The RDF and PDOS analyses reveal a progressive reduction in both C-C bond lengths and the full width at half-maximum of the primary RDF peak as diameter increases, ultimately converging to graphene-equivalent values. These structural changes reflect enhanced stiffness and phonon softening induced by radial strain in smaller-diameter CNTs that suppress thermal conductivity. This behavior is further illustrated by comparing the cumulative phonon density of states below 20 THz.

The thermal conductivity of DWCNTs with varying interlayer spacings has been calculated. The outer wall was fixed as a (9,9) SWCNT, while the inner wall was modulated among (4,4), (5,5), and (6,6) SWCNT configurations. Among these structures, (5,5)@9,9 exhibited the highest thermal conductivity, whereas (6,6)@9,9 demonstrated

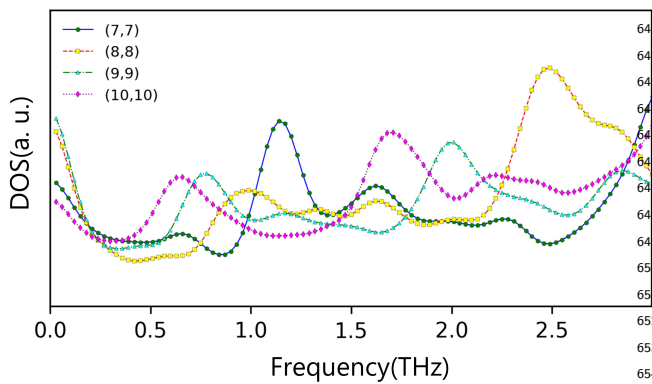


Figure 11: Blue shift of the phonon density of states peaks in carbon nanotubes with different diameters in the low-frequency region.

the lowest. Bond length analysis reveals pronounced radial deformation in the (6,6)@(9,9) DWCNT structure, characterized by inner wall compression and outer wall expansion. This structural distortion induces phonon hardening in the compressed inner wall and phonon softening in the expanded outer wall. Our analysis indicates that phonon softening constitutes the dominant mechanism, resulting in reduced thermal conductivity, whereas enhanced van der Waals interactions may provide a secondary contribution. The (5,5)@(9,9) structure, which has minimal structural deformation, exhibits the highest thermal conductivity among the investigated configurations. These findings suggest that strain engineering holds significant potential for effectively tuning the thermal conductivity of both single-walled and double-walled CNTs, enabling the design of materials for thermoelectric and thermal management applications.

Conflicts of interest

There are no conflicts to declare.

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Data Availability Statement

The potential model and example codes to reproduce the data of this manuscript are freely available at [GitHub](#).

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